



Brochure

NVIDIA ENTERPRISE AI REFERENCE ARCHITECTURE WITH HPE

Transforming data centers with flexible, scalable,
and powerful server GPU solutions



Redefining AI infrastructure with HPE and NVIDIA

Today's organizations are redefining what's possible with artificial intelligence (AI)—pushing boundaries in research, business, and innovation. At the center of this transformation stands the [NVIDIA AI Computing by HPE](#) portfolio, a collaboration that unlocks new potential in enterprise AI infrastructure. By fusing robust HPE compute, software, and storage platforms with the latest advances in NVIDIA GPUs and software in jointly-validated configurations, blueprints make it easier to deploy reliable, scalable, and high-performance AI solutions.

This reference architecture (RA) overview offers an NVIDIA-validated solution that scales from 4-32 nodes. HPE tailors and optimizes solution components for distinct use cases, environments and operational scale. Inside HPE ProLiant servers, NVIDIA® RTX PRO 6000 Blackwell Server Edition GPUs deliver an entry point into data center-scale AI, balancing performance, cost, and flexibility.

The collaboration between HPE and NVIDIA means this solution design is technically robust and backed by a collaborative ecosystem of support, management, and future-ready innovation. The result is a foundation that speeds time to insight for AI projects of any size, and enables enterprises to meet the evolving demands of data analytics and artificial intelligence.

Enterprise AI Reference Architecture with NVIDIA RTX PRO 6000 Blackwell Server Edition

Ideal use cases: This cluster architecture is designed for organizations seeking a flexible, scalable entry point into AI workloads at data center scale. It is particularly well-suited for AI inference and fine tuning for models less than 70B parameters, and a range of enterprise applications where modular expansion and balanced cost-performance are essential. The design is optimized for environments prioritizing rapid deployment, reliability, and efficient scaling, making it a strong fit for enterprise data centers launching new AI initiatives.

The NVIDIA Enterprise AI Reference Architecture (Enterprise RA) using RTX PRO 6000 Blackwell Server Edition GPUs are built to solve today's most challenging computational problems. This architecture employs a modular design, with each HPE ProLiant Compute DL380a Gen12 server equipped with up to eight RTX PRO 6000 Blackwell GPUs and the NVIDIA Spectrum-X™ Networking Platform to power next-generation AI use cases.



Key components:

- HPE ProLiant servers, each with up to 8 RTX PRO 6000 Blackwell Server Edition GPUs
- NVIDIA Spectrum-4 networking platform
- NVIDIA Connect-X-7 SuperNICs and DPUs
- NVIDIA AI Enterprise software

Architecture design:

- Composed of scalable units (SUs), each with 4 HPE ProLiant servers, allowing for rapid and flexible deployment at multiple scales
- Scales to 32 certified systems, for a total of 256 RTX PRO 6000 Blackwell GPUs in a fully tested system (8 SUs)
- Flexible, rail-optimized network architecture and support for custom rack layouts

Management and support:

- NVIDIA AI Enterprise, NVIDIA CUDA®, and NVIDIA Magnum IO™
- Hardware support via HPE fulfillment, software support via NVIDIA AI Enterprise subscription

Primary use cases:

- AI inference and fine-tuning for large models up to 70B parameters, powering copilots, assistants and agentic AI
- Hybrid AI and single precision HPC workloads requiring modular, scalable infrastructure for high-performance data processing and simulation
- Data analytics, visual computing and engineering simulation to accelerate insight, design, and decision-making

This Enterprise RA provides a robust and adaptable framework for organizations to innovate confidently, supported by the combined strengths of HPE and NVIDIA engineering, software, and support ecosystem.

Blueprints with NVIDIA: A platform for accelerated innovation

Reference architectures with NVIDIA represent a new era of enterprise-ready AI—designs to streamline AI adoption, accelerate deployment, and scale innovation across a wide range of industries and use cases. The resulting HPE AI solutions with NVIDIA provide a holistic environment where compute, software, storage, networking, and server GPU acceleration can be integrated to create an agile AI infrastructure.

Central to these architectures is a philosophy of modularity and scalability. Organizations can start with a single scalable unit and expand to hundreds of nodes as their needs evolve, thanks to a flexible design that adapts to both current and future demands. The platform integrates high-performance HPE ProLiant servers with the versatile RTX PRO 6000 Blackwell Server Edition GPUs, ideal for mainstream AI workloads and rapid experimentation.

Automation and orchestration are core to the HPE experience. With management solutions such as NVIDIA AI Enterprise, CUDA, and Magnum IO, the integrated platform efficiently deploys, monitors, and optimizes workloads. Enterprises gain access to a comprehensive suite of tools for AI inference, and data analytics, all supported by HPE and NVIDIA partners.

The portfolio of available designs integrates curated software with secure, validated infrastructure to provide an engineering-validated stack designed for today's AI workloads.

Collaboration is another hallmark of the enterprise AI design. The partnership between HPE and NVIDIA brings together expertise in solution engineering. The installation and deployment expertise from HPE gives enterprise clients the confidence to deploy AI at scale. The design's modular nature not only simplifies expansion, but also empowers organizations to integrate their AI workloads with broader IT initiatives.

For industries ranging from healthcare and life sciences to finance, manufacturing, and retail, HPE enterprise AI solutions with NVIDIA enable faster time to insight, greater operational efficiency, and the freedom to innovate. With engineering-validated designs, every enterprise, regardless of size or specialization, can harness the transformative power of AI.

Ultimately, HPE enterprise AI architectures with NVIDIA stand as a foundation for next-generation AI, designed for those determined to lead in an increasingly data-driven world. It is the key to unlocking new possibilities, automating complex processes, and driving breakthroughs across business.





Learn more at

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Why HPE for AI solutions

Many teams face familiar barriers, such as fragmented data, security concerns, and infrastructure that can't keep pace, making it difficult to move from AI pilot to production with confidence. HPE delivers AI solutions built for the realities of the enterprise, helping organizations progress from ambition to outcomes even when budgets, skills, and operational capacity are constrained. The HPE approach focuses on creating a strong, unified AI foundation designed for secure, sustainable, enterprise-scale deployment across hybrid environments.

HPE leads with experience and expertise to guide the workload-focused decision points that drive a right-sized design—what must run now versus later, which workloads require acceleration, and how to operationalize AI without adding complexity. With an extensive portfolio of AI solutions and services that emphasize trusted data pipelines, AI-tuned operations, and turnkey hybrid platforms, HPE helps speed time to value while preserving flexibility and control, enabling organizations to turn AI initiatives into repeatable, measurable business outcomes.

A foundation for the future of AI

This enterprise AI solution design sets a new standard for enterprise-ready, flexible, and future-proof AI deployment. With an architecture built for rapid entry into data center AI using HPE ProLiant servers with PCIe-based NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs inside, organizations have the freedom to innovate without compromise.

Whether pursuing AI inference and fine tuning, or data analytics, visual computing, and engineering simulation, the combination of engineering expertise from HPE and NVIDIA enables seamless scaling, simplified management, and sustained performance. As AI ambitions grow, this architecture can serve as a launching pad—backed by best-in-class support and a shared vision for what AI can achieve in the enterprise.

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